

```

1 • show databases;
2
3 • use employees;
4
5 • select * from employees;
6 • select * from departments;
7
8 #1
9 • SELECT
10     department_id, AVG(salary) average_salary
11 FROM
12     employees
13 GROUP BY department_id
14 HAVING AVG(salary) > 7000;
15
    
```

Result Grid
 

 Filter Rows:
 
 Export:
 
 Wrap Cell Content:

	department_id	average_salary
▶	57	14961.5238
	50	14798.8182
	63	14594.6429
	30	11759.1176
	59	14712.0909
	52	11996.2000
	35	14026.4400
	32	12344.7727
	71	12983.7500
	58	14476.3929



```
16
17
18
19 #2
20 • SELECT
21     *
22 FROM
23     employees
24 WHERE
25     first_name LIKE 'S%';
26
27
28
29
30
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	department_id	average_salary
►	57	14961.5238
	50	14798.8182
	63	14594.6429
	30	11759.1176
	59	14712.0909
	52	11996.2000
	35	14026.4400
	32	12344.7727
	71	12983.7500
	58	14476.3929

```

28
29
30
31 #3
32 • SELECT
33     e.employee_id, e.first_name, e.last_name, d.department_name
34 FROM
35     employees e
36     INNER JOIN
37     departments d ON e.employee_id = d.manager_id;
38
39
40
41
42
    
```

Result Grid

 Filter Rows:
 Export:

 Wrap Cell Content:

	department_id	average_salary
	58	14476.3929
	44	12239.4706
	61	11094.4815
	79	15089.0000
	60	15328.8148
	62	14297.1053
	69	16666.8462
	70	10482.0000
	67	12048.3889
	38	11210.8000

40

41

42

43 #4

44 • SELECT

45 employee_id, first_name, last_name

46 FROM

47 employees

48 WHERE

49 manager_id IS NULL;

50

51

52

53

54

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



employee_id	first_name	last_name
-------------	------------	-----------

```

52
53
54      #5
55 •   SELECT
56         e.employee_id, e.first_name, e.last_name, d.department_name
57   FROM
58         employees e
59         LEFT JOIN
60         departments d ON e.department_id = d.department_id;
61
62
63
64
65
66

```

Result Grid

 Filter Rows: _____
 Export:
 Wrap Cell Content:
 Fetch rows:

	employee_id	first_name	last_name	department_name
▶	100	Sarah	Chen	Archivist
	101	Victor	Mcgee	Clinical scientist, histocompatibility and immunog...
	102	Brittney	Mccormick	Advice worker
	103	Taylor	James	Engineer, broadcasting (operations)
	104	James	Anderson	Manufacturing engineer
	105	James	Schmidt	Advertising art director
	106	Stephanie	Sharp	Communications engineer
	107	Elizabeth	Carter	Engineer, maintenance
	108	Martin	Mckinney	Veterinary surgeon
	109	Jo	Brown	Ship broker



62

63

64

65 #6

66 • SELECT

67 YEAR(hire_date) hire_date, COUNT(employee_id) employee_count

68 FROM

69 employees

70 GROUP BY YEAR(hire_date)

71 ORDER BY hire_date;

72

--

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



	hire_date	employee_count
►	2014	31
	2015	104
	2016	97
	2017	114
	2018	104
	2019	102
	2020	96
	2021	106
	2022	91
	2023	77
	2024	78



```
72
73      #7
74 •    SELECT
75         job_id, COUNT(employee_id) employee_count
76     FROM
77         employees
78     GROUP BY job_id;
79
80
81
82
83
```

Result Grid

Filter Rows:

Export:



Wrap Cell Content:



	job_id	employee_count
▶	IT_PROG	142
	AD_VP	141
	FI_ACCOUNT	128
	PU_CLERK	158
	AD PRES	155
	PU_MAN	137
	FI_MGR	139



```
80
81
82      #8
83 •   SELECT
84         employee_id, first_name, last_name, salary
85     FROM
86         employees
87     ORDER BY salary DESC
88     LIMIT 5;
```

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



Fetch rows:



	employee_id	first_name	last_name	salary
▶	447	Kristine	Shea	24958
	203	Alicia	Kidd	24932
	933	Diane	Griffith	24927
	657	Claire	Lowery	24918
	772	Matthew	Bell	24894



```

94
95 #9
96 • SELECT
97     employee_id, first_name, last_name, salary
98 FROM
99     employees
100 WHERE
101     MOD(salary, 2) = 0
102 ORDER BY salary;
103

```

Result Grid Filter Rows: Export: Wrap Cell Content:

	employee_id	first_name	last_name	salary
▶	320	Ryan	Melton	2616
	347	Jacob	Santana	2624
	291	Laurie	Goodwin	2660
	1092	Jon	Gonzalez	2688
	803	Laura	Cochran	2762
	333	Jeanette	Nguyen	2770
	719	Jeffrey	King	2872
	163	Sheila	Ramirez	2904
	118	Kimberly	Erickson	2958
	317	Lauren	Estrada	2998
	394	Carl	Roy	3008
	949	Jodi	Parker	3082
	895	Scott	Horne	3196
	526	Matthew	Baker	3336
	764	Kyle	Evans	3354
	176	Sandra	Reid	3360

```
104
105      #10
106 •   SELECT DISTINCT
107         d.department_name
108   FROM
109         departments d
110        INNER JOIN
111         employees e ON d.department_id = e.department_id
112   ORDER BY d.department_name;
113
```

	department_name
▶	Advertising art director
	Advertising copywriter
	Advice worker
	Animal nutritionist
	Applications developer
	Archivist
	Arts development officer
	Broadcast journalist
	Chartered accountant
	Chartered legal executive (England and Wales)
	Chief Financial Officer
	Clinical scientist, histocompatibility and immuno...
	Communications engineer
	Dancer
	Data processing manager
	Early years teacher
	...

```

113
114      #11
115 •   SELECT
116         employee_id, first_name, last_name, hire_date
117     FROM
118         employees
119     WHERE
120         DAY(hire_date) % 2 = 1
121     ORDER BY hire_date;
122

```

Result Grid   Filter Rows: | Export:  | Wrap Cell Content: 

	employee_id	first_name	last_name	hire_date
▶	1022	Cristina	Miller	2014-08-13
	519	Katherine	Mitchell	2014-08-31
	791	Lindsay	Adams	2014-09-15
	281	Steven	Russell	2014-09-17
	867	Regina	Werner	2014-09-19
	979	Benjamin	Joseph	2014-09-19
	741	Peter	Boone	2014-10-01
	354	Debra	Atkinson	2014-10-15
	1050	Samantha	Case	2014-11-07
	1080	Debbie	Sanchez	2014-11-09
	864	Ronald	Stewart	2014-11-27
	602	Breanna	Herman	2014-12-07
	665	Richard	Anderson	2014-12-07
	826	John	Horton	2014-12-13
	971	Michael	Williamson	2014-12-17
	643	Audrey	Bowen	2014-12-23
	687	...	...	...



123

124 #12

125 • SELECT DISTINCT

126 department_name

127 FROM

128 departments;

129

130

131

132

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



department_name

► Engineer, broadcasting (operations)

Advertising copywriter

Engineer, maintenance

Mining engineer

Research scientist (maths)

Communications engineer

Location manager

Geographical information systems officer

Naval architect

Tourism officer

Data processing manager

Outdoor activities/education manager

Arts development officer

Early years teacher

Applications developer

Surveyor, planning and development



```

130      #13
131 •    SELECT
132         e.employee_id AS employee_id,
133         e.first_name AS employee_first_name,
134         e.last_name AS employee_last_name,
135         m.employee_id AS manager_id,
136         m.first_name AS manager_first_name,
137         m.last_name AS manager_last_name
138     FROM
139         employees e
140         LEFT JOIN
141         employees m ON e.manager_id = m.employee_id;
142

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

	employee_id	employee_first_name	employee_last_name	manager_id	manager_first_name	manager_last_name
▶	100	Sarah	Chen	113	Elizabeth	Carney
	101	Victor	Mcgee	140	Robert	Sanchez
	102	Brittney	Mccormick	105	James	Schmidt
	103	Taylor	James	139	Morgan	Keith
	104	James	Anderson	108	Martin	Mckinney
	105	James	Schmidt	141	Donna	Johnson
	106	Stephanie	Sharp	122	Jessica	Horton
	107	Elizabeth	Carter	118	Kimberly	Erickson
	108	Martin	Mckinney	105	James	Schmidt
	109	Jo	Brown	138	Ashley	Phillips
	110	Lisa	Miles	148	Juan	Lewis
	111	Richard	White	136	Laura	Cook



```

142
143
144      #14
145  ●   SELECT
146      e.employee_id,
147      e.first_name,
148      e.last_name,
149      e.salary,
150      e.department_id
151  FROM
152      employees e
153      JOIN
154      (SELECT
155          department_id, AVG(salary) AS avg_salary
156      FROM
157          employees
158      GROUP BY department_id) d_avg ON e.department_id = d_avg.department_id
159  WHERE
160      e.salary > d_avg.avg_salary;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	employee_id	first_name	last_name	salary	department_id
▶	101	Victor	Mcgee	20947	50
	102	Brittney	Mccormick	15367	63
	105	James	Schmidt	14951	52
	106	Stephanie	Sharp	21032	35
	109	Jo	Brown	21101	58
	111	Richard	White	13928	61
	112	Kathleen	Becker	15833	79
	113	Elizabeth	Carney	24213	60
	114	April	Hill	16076	62
	115	Angela	Ayala	17075	69



```

161
162
163
164      #15
165 •   SELECT
166         d.department_name, COUNT(e.employee_id) AS employee_count
167     FROM
168         departments d
169         LEFT JOIN
170         employees e ON d.department_id = e.department_id
171     GROUP BY d.department_name;
172
173
174
175
176
177
178

```

Result Grid | | Filter Rows: | Export: | Wrap Cell Content:

	department_name	employee_count
▶	Engineer, broadcasting (operations)	17
	Advertising copywriter	19
	Engineer, maintenance	22
	Mining engineer	24
	Research scientist (maths)	17
	Communications engineer	25
	Location manager	19
	Geographical information systems officer	25
	Naval architect	15
	Tourism officer	24



```

173
174 #16 #i create a view as you can see i update a query.
175 create view ss as
176 SELECT
177     e.employee_id,
178     e.first_name,
179     e.last_name,
180     e.salary,
181     e.department_id
182 FROM
183     employees e
184     JOIN
185     (SELECT
186         department_id, AVG(salary) AS avg_salary
187     FROM
188         employees
189     GROUP BY department_id) d_avg ON e.department_id = d_avg.department_id
190 WHERE
191     e.salary > d_avg.avg_salary;
192 • select * from ss;
...

```

Result Grid Filter Rows: Export: Wrap Cell Content:

	employee_id	first_name	last_name	salary	department_id
▶	101	Victor	Mcgee	20947	50
	102	Brittney	Mccormick	15367	63
	105	James	Schmidt	14951	52
	106	Stephanie	Sharp	21032	35
	109	Jo	Brown	21101	58
	111	Richard	White	13928	61
	112	Kathleen	Becker	15833	79
	113	Elizabeth	Carney	24213	60
	114	Anril	Hill	16076	62